25901884 **Reg. No.:** .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)
Third Semester
Value Addition Course (VAC) – MG3VACBCH200 – MICROPLASTICS AND ENVIRONMENT
(2024 Admission Onwards)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Multiple Choice Questions

Multiple Choice Questions
Answer all questions
Each question carries 1 marks

1. Which statement best explains why rivers are important in the transport of microplastics?
   Answer: c) They carry microplastics from land to oceans.
2. Which best explains why microplastics undergo biomagnification?
   Answer: d) They accumulate in tissues and are transferred through the food chain.
3. The impact on soil fauna by the ingestion of microplastics are
   Answer: c) Physical harm, reduced growth.
4. Indicate the reason why it is difficult to avoid microplastic contamination in foods like honey and sugar.
   Answer: c) Microplastics are widespread in the environment and can enter food at many stages.
5. Which natural system is disturbed by the presence of microplastics in oceans?
   Answer: b) Biological carbon pump.

Part B

Fill In The Blank Questions
Answer all questions
Each question carries 2 marks

6. The type of radiation that degrades plastics is ----------
   Answer: Ultraviolet (UV) radiation.
7. Microplastics that result from the breakdown of larger plastic items are known as ----------microplastics.

Answer : secondary

8. Economic losses in ----------are linked to polluted beaches and degraded marine environments.
   Answer :tourism
9. Microplastics in the digestive tract can cause ----------of the gastrointestinal lining.
   Answer : inflammation
10. Guidelines for industries are created to reduce----------and promote ------- management.
    Answer : plastic pollution, waste

Part C
Short Answer Questions

Answer any 5 questions. Each question carries 3 marks.

11. Define microplastic degradation. Give two examples of environmental factors that influence the degradation. [K] / [CO1]
    Answer: Definition of microplastic degradation **1 mark**
    Any two environmental factors influencing degradation **2 marks**
12. Define bioaccumulation and explain the difference between bioaccumulation and biomagnification. [K] / [CO1]
    Answer : Definition of bioaccumulation. **(1 mark)**
    Definition/ of biomagnification. **(1 mark)**
    Difference: **(1 mark)**
13. Give an outline regarding the sources of Microbeads. [U] / [CO1]
    Answer: Explain any three Sources **(one mark each)**
14. Label the microplastic exposure pathways for fish and shellfish, and explain the observed effects. [K] / [CO2]
    Answer: Explanation **-3marks**
15. Why is bottled water often found to contain higher levels of microplastics than tap water? [An] / [CO3]
    Answer: Any three points **(one mark each)**

16. Explain the impacts of microplastics on the cryosphere. How is albedo influenced by microplastics? [U] / [CO4]
    Answer : Explanation of the impacts of microplastics on the cryosphere **2marks**
    Effect of Albedo- 1mark
17. Recall the effects of microplastics on habitat destruction. [K] / [CO3]
    Answer: Any three effects **(Three mark each)**

[3 × 5 = 15]

Part D
Short Essay Questions

Answer any 4 questions. Each question carries 5 marks.

18. Explain how sewage sludge and biosolids used as fertilizers contribute to soil microplastic pollution. [K] / [CO1]

Answer: Source of microplastics in sewage sludge – **1 mark**
Use of biosolids as fertilizer – **1 mark**
Entry of microplastics into soil – **1 mark**

Accumulation due to repeated application – **1 mark**
Effects on soil and plants/food chain – **1 mark**

19. Recall the role of industrial abrasives and pre-production pellets as microplastics. [U] / [CO1]

Answer: Industrial abrasives as microplastics – **1 mark**
Release of abrasives into the environment – **1 mark**
Definition of pre-production pellets (nurdles) – **1 mark**
Pellet loss during handling/transport – **1 mark**
Environmental pollution caused by both – **1 mark**

20. Relate the key effects of microplastics on freshwater ecosystems to broader ecological and human impacts. [U] / [CO2]

Answer: **I**ngestion by freshwater organisms – **1 mark**
Effects on growth, reproduction, or biodiversity – **1 mark**
Transport of toxic chemicals/pathogens – **1 mark**
Bioaccumulation/food chain transfer – **1 mark**
Human exposure and health impacts – **1 mark**

21. Describe how scientific limitations contribute to uncertainty in microplastic health research. [U] / [CO3]
Answer: Any 5 points **(one mark each)**

22. Outline the primary exposure route of microplastics to humans. [U] / [CO1]
Answer: Explain any 5 primary exposure routs **(one mark each)**

23. Compare the physical and biological methods of microplastic removal strategy.
Answer: Physical methods **-1.5 marks**
Biological methods – **1.5 marks**
Comparison **-2 Marks**
[U] / [CO4]